

Appliances Control With Yoda Module And Arduino

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This project is meant to control electrical appliances like fans and lights. You can configure two Yoda modules, one as receiver and the other as transmitter, to control relays that switch on fans and lights. The block diagram for the project is shown in Fig. 1.

Transmitter

The transmitter comprises Arduino Nano (Board1), Yoda module (YM1), Yoda adaptor (YA1), two resistors (R1 and R2), and two on/off switches (S1 and S2), as shown in Fig. 2.

Yoda module (YM1) acts as a Zigbee coordinator RF module. For connecting the module to Arduino, you would require an adaptor. The Yoda module mounted on its adaptor is shown in Fig. 3.

The transmitter circuit (shown in Fig. 2) wired on a breadboard is shown in Fig. 4 for reference. Connect one end of the micro USB cable to Arduino Nano board and the other to your PC/laptop to power on the transmitter circuit and to upload the Arduino code.

Before using the transmitter circuit, you need to program the Arduino Nano with Arduino code (remote_tx.ino) and the Yoda module to be configured as explained later.

The use of Yoda module and adaptor is explained in their manual. Software configurations are explained in Yoda Command Set document. Both these can be found at <https://drive.google.com/open?id=1fa8ruNQ7Y2IKUxUFJQBslBjvWKn6zh5>

Receiver

The receiver is built around Arduino Nano (Board2), Yoda module (YM2), Yoda adaptor (YA2), 2-channel relay module (RM1), and 5V DC power adaptor. Here, the Yoda module acts as an end-node module. The receiver circuit (shown in Fig. 5) wired on a

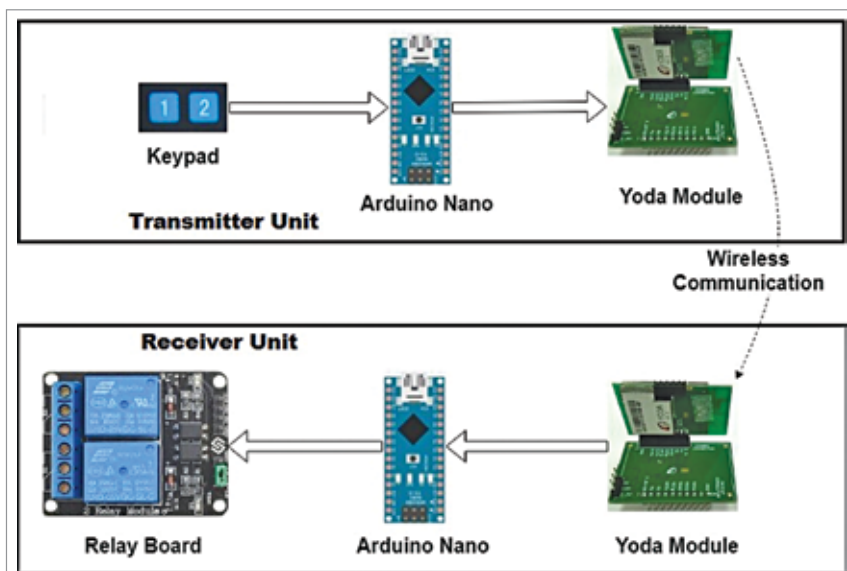


Fig. 1: Block diagram of transmitter and receiver in the project

breadboard is shown in Fig. 6.

As Arduino Nano board (Board2) cannot provide sufficient current to drive a relay, a 5V DC power adaptor is used for powering the relay module/board (RM1). To use 5V DC power adaptor with the circuit, you may use a DC jack, as shown in Fig. 7(a), or a DC socket with twin wire, as shown in Fig. 7(b). To use DC jack in breadboard, solder two wires as shown in Fig. 8.

A black colour male-to-male Dupont jumper wire can be soldered on the negative terminal of DC jack for connection to breadboard and a red colour Dupont wire can be soldered to positive terminal for connection to

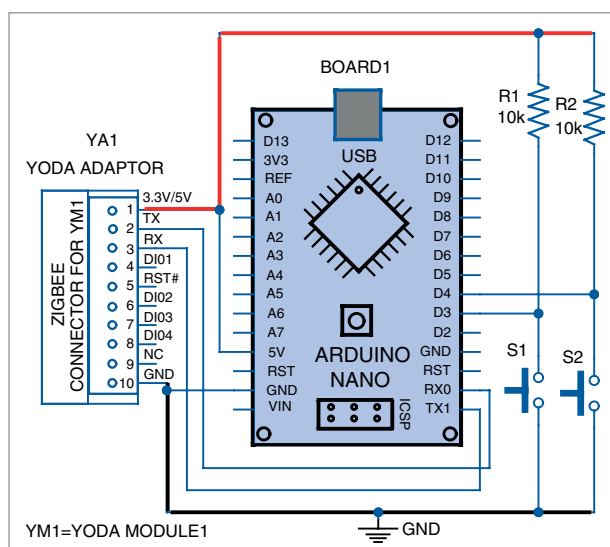


Fig. 2: Transmitter circuit

the relay board.

Connect one end of the micro USB cable to Arduino Nano board (Board2) and the other to PC/desktop to power on the receiver circuit and upload the Arduino code (relay_rx.ino). You can also connect a 5V, 1.5A DC power